

Manipulate Open Data : Renewable Power Plant

A rectangular badge with a dark grey background and white text. The word "License" is on the left and "CC BY-SA 4.0" is on the right.

Figure 1: License: CC BY-SA 4.0

First download csv data.

Data discovery

in terms of power

Use `pandas` to manipulate data ? Number of plants ? Look at the statistical description of the data ? Take a closer look at `electrical_power` and at all the quartile ? Draw an histogram to have an idea of the distribution of how the power plants power looks like ? In particular for the plants that have an `electrical_power` less than a threshold defined by the 90 decile.

in terms of localisation.

Implement a function that compute the haversine distance between to plants. or use scikit learn function

```
def haversine(lat1, lon1, lat2, lon2):  
    ...
```

Then write a code that search the plant the nearest of one of the row of your dataframe. At the end, you have something like:

Nearest plan to "Saint-Flour" is "Trézioux"

export json

1. Export several rows of the dataframe as dict in json format.
2. Export the N first rows of the dataframe in a file in bulk json format

Crud Manipulation

1. Create an index called `power__plant`
2. Index documents using file
3. Read the document to check the content of the document
4. Update a field in this document
5. Reread the document to ensure that the field has been updated.

6. Delete the document
7. Delete the index

Mapping definition

1. Create an index called `power_plant`
2. look at the current mapping of your index
3. index document
4. look at the mapping.
5. delete the index
6. Create the right mapping for coordinates

Bulk operations

1. Create one `power_plant` index with the right mapping
2. Index all power plants using the `json` you previously exported.
3. Try to build a request that allows to do the same as previously (find the nearest powerplant from another one)
4. Build a dashboard that localize on a map all power plants.

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